How-To Library

Informatica

Address Verification Best Practices for Taiwan

© Copyright Informatica LLC 1993, 2021. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

This document tells you how to connect the elements in a Taiwan address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in Taiwan.

Supported Versions

- Informatica Address Verification 3.0 and 4.0
- Informatica Address Verification (On-Premises) 5.x

Table of Contents

Introduction. . . . 2
Process Modes. . . . 2
Character Sets. . . . 3
Address Format and Address Structure. . . . 3
Key Address Elements. . . . 4
Mapping Input Address Data to Address Elements. . . . 4
Address Enrichments. . . . 6
Address Verification Tips and Best Practices. . . . 7
Glossary of Address Terms. . . . 7
Frequently Asked Questions. . . . 8
Reference Address Databases. . . . 9
Additional Documentation Resources. . . . 9

Introduction

Informatica Address Verification can validate and enhance addresses from Taiwan. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Taiwan address data.

To verify the quality of an input address, Address Verification compares the address to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies can determine the elements that you need.

An address in Taiwan can contain different types of information, including dependent locality information and secondary street information. You must map each item of address information to an appropriate element in Address Verification.

Process Modes

Informatica Address Verification supports the following process modes for Taiwan addresses:

- BATCH

- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE
- PARSE

Character Sets

Informatica Address Verification can verify Taiwan addresses in traditional Chinese and Latin scripts.

By default, Address Verification returns the verified Taiwan addresses in the traditional Chinese script. Address Verification returns a transliterated version of the Chinese address in the Latin script.

The Taiwan postal authorities recognize different structures for an address in each script.

Address Format and Address Structure

You improve the performance of the Address Verification engine when you define the input address in the format that the postal service expects.

The address structure that you define depends on the character set that you use. As a rule, a Taiwan address in the Latin script presents the address elements in the opposite order to a Taiwan address in the traditional Chinese script.

Address Structure for Latin Characters

When you create an address in Latin characters, the postal service expects the following multi-line structure:

```
Line 1: RECIPIENT
Line 2: [BUILDING INFORMATION]
Line 3: [SUB-BUILDING INFORMATION]
Line 4: [DEPENDENT STREET INFORMATION]
Line 5: [HOUSE/BUILDING NUMBER] STREET
Line 6: [DEPENDENT LOCALITY]
Line 7: LOCALITY POSTAL CODE
Line 8: [COUNTRY]
```

The following sample address demonstrates the structure:

```
55 Jinshan S. Rd., Sec. 2
Da'an Dist.
Taipei City 10603
TAIWAN
```

Address Structure for Traditional Chinese Characters

A Taiwan address in the native script lists all address elements on a single line.

You can validate a Taiwan address in the traditional Chinese script as a single string in a Formatted Address Line element.

When you format an input address, enter the elements in the address in the following order:

```
Postal Code, Locality, Dependent Locality, Street, Dependent Street, House or Building Number,
Building Name, Sub-Building Name
```

The following sample address illustrates the address in the native script:

```
10603 臺北市大安區金山南路２段 55 號
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Street
- House Number
- Postal Code

Mapping Input Address Data to Address Elements

To verify an address, map the input address data elements to the appropriate elements in Address Verification.

The following table shows how the above Chinese input address data maps to the Address Verification address elements:

Information Type	Address Elements	Sample Address Information
Postal Code	PostalCode	10603
City	Locality_1	臺北市
District	Locality_2	大安區
Village and Neighborhood	Locality_3	[Not used in this address]
Street Direction, Road, and Section	Street_1	金山南路 2 段
Dependent street, for example Lane	Street_2	[Not used in this address]
Double-dependent street, for example Alley	Street_3	[Not used in this address]
House or Building Number	House Number	55 號
Sub-building information	Sub-Building	[Not used in this address]

Input and Result XML in Address Verification (On-Premises)

The following example shows the address elements in the input and result XML in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
   <AddressElements>
    <Country Item="1" Type="NAME">TWN</Country>
   </AddressElements>
   <AddressLines>
    <FormattedAddressLine Line="1">10603 臺北市大安區金山南路 2 段 55 號 </FormattedAddressLine>
   </AddressLines>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<Result  ProcessStatus="V4"
               ModeUsed="BATCH"
               Count="1"
               CountOverflow="NO"
               CountryISO3="TWN"
```

```
                PreferredScript="DATABASE"
                PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
                MailabilityScore="5"
                ResultPercentage="100.00"
                ElementResultStatus="F0FF00F0F000000000E0"
                ElementInputStatus="606600606000000000060"
                AddressResolutionCode="00000000000000000000"
                ExtElementStatus="00550050000000000000"
                ElementRelevance="10110010100000000010"
                AddressType="S"
                LanguageISO3="CHN">
   <AddressElements>
              <Country Type="NAME_EN" Item="1">TAIWAN</Country>
              <Locality Item="1">臺北市</Locality>
              <Locality Item="2">大安區</Locality>
              <PostalCode Item="1">10603</PostalCode>
              <Street Item="1">金山南路, 2段</Street>
              <Number Item="1">55號</Number>
   </AddressElements>
   <AddressLines>
     <DeliveryAddressLine Line="1">金山南路2段55號</DeliveryAddressLine>
     <CountrySpecificLocalityLine Line="1">10603臺北市大安區</CountrySpecificLocalityLine>
     <FormattedAddressLine Line="1">10603臺北市大安區金山南路2段55號</FormattedAddressLine>
   </AddressLines>
     <AddressComplete>10603臺北市大安區金山南路2段55號</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Taiwan address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
 <soap:Header/>
 <soap:Body>
   <v:Process>
     <v:login>****</v:login>
     <v:password>****</v:password>
     <v:parameters>
       <v:ProcessMode>BATCH</v:ProcessMode>
       <v:ServiceParameters>
       </v:ServiceParameters>
       <v:ValidationParameters>
       </v:ValidationParameters>
     </v:parameters>
     <v:addresses>
       <v:Address>
         <v:Country>
           <v:string>TWN</v:string>
         </v:Country>
           <v:AddressComplete>10603臺北市大安區金山南路2段55號</v:AddressComplete>
       </v:Address>
     </v:addresses>
   </v:Process>
 </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
```

```
                <Result>
                   <ProcessStatus>V4</ProcessStatus>
                   <CountryISO3>TWN</CountryISO3>
                   <ResultDataSet>
                      <ResultData>
                       <ResultNumber>1</ResultNumber>
                       <MailabilityScore>5</MailabilityScore>
                       <ResultPercentage>100.00</ResultPercentage>
                       <ElementInputStatus>6066006060000000060</ElementInputStatus>
                       <ElementResultStatus>F0FF00F0F000000000E0</ElementResultStatus>
                       <ElementRelevance>10110010100000000010</ElementRelevance>
                       <ExtElementStatus>00550050000000000000</ExtElementStatus>
                       <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                         <AddressType>S</AddressType>
                         <LanguageISO3>CHN</LanguageISO3>
                         <Address>
                            <Street>
                               <string>金山南路, 2段</string>
                            </Street>
                            <HouseNumber>
                               <string>55 號</string>
                            </HouseNumber>
                            <Locality>
                               <string>臺北市</string>
                               <string>大安區</string>
                            </Locality>
                            <PostalCode>
                               <string>10603</string>
                            </PostalCode>
                            <Country>
                               <string>TAIWAN</string>
                            </Country>
                            <DeliveryAddressLines>
                               <string>金山南路 2 段 55 號</string>
                            </DeliveryAddressLines>
                            <CountrySpecificLocalityLine>
                               <string>10603 臺北市大安區</string>
                            </CountrySpecificLocalityLine>
                            <FormattedAddress>
                               <string>台北 10603 臺北市大安區金山南路 2 段 55 號</string>
                            </FormattedAddress>
                         <AddressComplete>台北 10603 臺北市大安區金山南路 2 段 55 號</AddressComplete>
                         </Address>
                      </ResultData>
                   </ResultDataSet>
                </Result>
             </Results>
          </ProcessResult>
       </ProcessResponse>
    </soap:Body>
 </soap:Envelope>
```

Address Enrichments

You can configure Informatica Address Verification to add the following enrichment to verified Taiwan addresses:

Enrichment	Database	Description
Standard Geocoding	TWN5GC.MD	Retrieves the geocoordinates for valid Taiwan addresses. Geocoordinates are accurate to the locality level.

Address Verification Tips and Best Practices

Note the following points when you verify Taiwan addresses:

- The input address elements in a traditional Chinese address must include indicators that identify the address elements. For example, the City data in an input address must include the character **市**, which indicates a city. The indicators follow the data elements that they describe.

  Note: Do not include a postal code identifier in the postal code input data.

  If you do not specify indicators in a Taiwan address, Address Verification rejects that address and does not return quality results for the input address.
- Do not include commas or spaces between the characters in a traditional Chinese address.
- A building number might represent a building complex. The address might identify the building complex and a building or block within the complex. In this case, use an additional building element to identify the building or block within the complex.
- Address Verification supports verification of Taiwan addresses up to the house number level.
- Address Verification does not verify contact data. Do not include contact data in an unfielded input address, for example on a formatted address line. You can include contact data in an input address if you define the contact information in a Contact address element. Address Verification passes the input contact data to the output address.

Glossary of Address Terms

Taiwan addresses in traditional Chinese contain descriptive terms for one or more items of address information. You must add one or more terms to improve the quality of addresses. For example, append 市 to a city name.

Add the descriptive terms after each item of address information.

The following table lists common descriptive terms and the English equivalents:

Simplified Chinese Version	English Version and Usage
市	Indicates a city. Assign to a Locality_1 element.
縣	Indicates a county. Assign to a Locality_1 element.
區	Indicates a district. Assign to a Locality_2 element.
鎮 or 鄉	Indicates a township or a rural division. Assign to a Locality_2 element.
村	Indicates a village. Assign to a Locality_3 element.
鄰 or 里	Indicates a neighborhood. Assign to a Locality_3 element.
路	Indicates a road. Assign to a Street_1 element.

Simplified Chinese Version	English Version and Usage
街	Indicates a street. Assign to a Street_1 element.
大道	Indicates a boulevard. Assign to a Street_1 element. You can abbreviate the element to 道.
段	Indicates a section. Assign to a Street_1 element.
巷	Indicates a lane. Assign to a Street_2 element.
弄	Indicates an alley. Assign to a Street_3 element if Street_2 data is present. Otherwise, assign to a Street_2 element.
號	Indicates a number. Assign to a House Number element.
樓	Indicates a floor. Assign to a Sub-building_1 element.
室	Indicates a room. Assign to a Sub-building_2 element.
東 西 南 北	Indicates an east direction. Indicates a west direction. Indicates a south direction. Indicates a north direction. Assign to a Street Directional element.
之 or 附	Indicates a separator between the characters. Assign as a connector between number elements.

Frequently Asked Questions

Why does Formatted Address Line output appear on a single line for Taiwan?

According to the postal regulations of the country, a Taiwan output address in the native language lists address elements in a single line.

For example, enter the following address:

```
Address_Ln1: 10603 臺北市大安區金山南路 2 段 55 號
Country: TWN
```

Address Verification returns the following output address in a single line:

```
<AddressElements>
     <Country Type="NAME_EN" Item="1">TAIWAN</Country>
     <Locality Item="1">臺北市</Locality>
     <Locality Item="2">大安區</Locality>
     <PostalCode Item="1">10603</PostalCode>
```

```
      <Street Item="1">金山南路, 2段</Street>
      <Number Item="1">55 號</Number>
</AddressElements>
<AddressLines>
  <DeliveryAddressLine Line="1">金山南路 2 段 55 號</DeliveryAddressLine>
  <CountrySpecificLocalityLine Line="1">10603 臺北市大安區</CountrySpecificLocalityLine>
  <FormattedAddressLine Line="1">10603 臺北市大安區金山南路 2 段 55 號</FormattedAddressLine>
</AddressLines>
  <AddressComplete>10603 臺北市大安區金山南路 2 段 55 號</AddressComplete>
```

What character scripts does Address Verification support?

From version 5.6.0 onward you can validate Taiwan addresses in the Mandarin Traditional Chinese and Latin scripts.

Where does the postal code information appear in a Taiwan address?

For an address in the traditional Chinese script, which has a single line address structure, the address begins with a postal code.

For example, the following traditional Chinese address illustrates that the address begins with the postal code:

```
10603 臺北市大安區金山南路 2 段 55 號
```

For an address in the Latin script, which has a multi-line address structure, postal code appears on the last line of the address.

For example, the following sample address demonstrates the address in the Latin script:

```
55 Jinshan S. Rd., Sec. 2
Da'an Dist.
Taipei City 10603
```

Does Address Verification support the postal code (郵編) indicator?

No, the current version of Address Verification does not support the postal code indicator.

Reference Address Databases

Informatica updates the Taiwan reference address databases irregularly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the UpData newsletter from Informatica.

Informatica Address Verification supports the following reference databases for Taiwan addresses:

- `TWN5BI.MD`. Batch and interactive modes.
- `TWN5FC.MD`. Fast completion mode.
- `TWN5GC.MD`. Standard geocoding.

Note: You download databases only for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

Author

Shahani Natalia Mendonca

Acknowledgements

The author would like to thank Amarpal Kaur Sohi, David Handy, Markus Loechel, Farhan Ashraf, and Xiayi Zhang for their assistance.